

FRESHNESS CANON

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Abstract

FRESHNESS is deterministic caching unchanged sources produce identical outputs, skip the work, trust the cache.

hadleylab.org Governed Research. Every claim cited.

Contents

- **DETERMINISTIC_OUTPUTS** the compiler must produce identical output for unchanged sources; the cache encodes this determinism.
- **SKIP_WHEN_CLEAN** when sources are unchanged, the work is skipped and the cached output is served; no redundant recompilation.
- **FRESHNESS_IS_SERVICE** FRESHNESS is a TOOLCHAIN service whose function is CACHE; the primitive is INTEL (what the compiler knows about source state). ## Constraints

MUST: Source dependency rules are declared in FRESHNESS

MUST: TOOLCHAIN hash change invalidates entire cache

MUST: CI always runs --no-cache (correctness is the goal)

MUST: Cache is a runtime artifact at ~/.canonic/.build

MUST: MONITORING deep health reports compile freshness

MUST: Freshness decisions emit structured output (JSON)

MUST NOT: Skip validation after incremental build 255

MUST NOT: Skip GC pass orphan cleanup always executes

MUST NOT: Cache stale outputs missing output files for

FRESHNESS / CANON / TOOLCHAIN